

SSBMF Status Update: Preprocessing Comparison, Cohort Indicator, and Final Configuration (May 27, 2026)

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1. Objectives and Decisions Addressed Since Last Update

This report covers three questions that arose from the May 8 update and have now been resolved through a comprehensive 18-configuration benchmark on merged TCGA_PAAD + CPTAC training data ($n = 273$, 5-fold CV K-selection with biological floor $K \geq 3$, evaluated on 5 external cohorts).

- **Which preprocessing strategy is compatible with merged multi-platform training?** Five strategies compared: joint quantile normalization + rank transform (prior standard), per-platform z-standardization, joint QN without rank transform, joint z-standardization, and log-transform only.
- **Does the cohort indicator column in L improve generalization?** The cohort indicator absorbs the RNA-seq vs. proteomics platform offset at the model level (as opposed to removing it in preprocessing). Evaluated for both LB and YFB across preprocessing strategies.
- **LB vs. YFB for the final manuscript configuration?** LB ($\eta = L\beta$) requires an OLS approximation to score new subjects ($\hat{L}_{\text{new}} \approx Y_{\text{new}} F (F^\top F)^{-1}$). YFB ($\eta = (YF)\beta$) scores new subjects exactly ($\eta_{\text{new}} = Y_{\text{new}} F \beta$) with no approximation.

2. Key Results: 18-Configuration Benchmark

Table 1: 18-configuration benchmark sorted by model form (LB then YFB). K_{eff} = active factors ($|\hat{\beta}_k| > 0.001$) after convergence. *Collapsed* = $\hat{\beta} = 0$ for all factors.

ID	Model	Preprocessing	Cohort	K	K_{eff}	Mean C
M1	LB	Joint QN+rank	No	6	1	0.599
M2	LB	Joint QN+rank	Yes	6	3	0.619
M3	LB	Per-plat z-std	No	3	1	0.620
M4	LB	Per-plat z-std	Yes	3	1	0.624
M7	LB	Joint QN	No	6	1	0.598
M8	LB	Joint QN	Yes	6	1	0.615
M9	LB	Joint z-std	No	3	0	0.545 (<i>collapsed</i>)
M10	LB	Joint z-std	Yes	3	0	0.555 (<i>collapsed</i>)
M11	LB	Log only	No	3	0	0.533 (<i>collapsed</i>)
M12	LB	Log only	Yes	3	0	0.531 (<i>collapsed</i>)
M5	YFB	Per-plat z-std	No	3	2	0.626
M6	YFB	Per-plat z-std	Yes	3	2	0.613
M13	YFB	Joint QN	No	5	0	0.540 (<i>collapsed</i>)
M14	YFB	Joint QN	Yes	5	0	0.529 (<i>collapsed</i>)
M15	YFB	Joint z-std	No	3	0	0.562 (<i>collapsed</i>)
M16	YFB	Joint z-std	Yes	3	0	0.543 (<i>collapsed</i>)
M17	YFB	Log only	No	3	0	0.529 (<i>collapsed</i>)
M18	YFB	Log only	Yes	3	0	0.532 (<i>collapsed</i>)

Critical finding: 10 of 12 non-per-platform configurations produce $\hat{\beta} = 0$ for all factors ($K_{\text{eff}} = 0$). The only viable preprocessing for merged RNA-seq + proteomics training is per-platform z-standardization applied before merging.

3. Preprocessing: Why Per-Platform Z-Standardization Is Required

When TCGA_PAAD (RNA-seq) and CPTAC (proteomics) expression matrices are stacked without per-platform normalization, the top singular vectors of the merged Y matrix capture the platform contrast, not biological variation. The CAVI survival update receives near-zero signal in these directions, so $\hat{\beta} \rightarrow 0$ regardless of model form or further normalization. Rank transform (M13/M14 vs. joint QN): removing the rank transform from joint-QN data does *not* prevent YFB collapse, ruling out rank transform as the mechanism. The root cause is the platform-dominated SVD shared by all joint preprocessing strategies.

Per-platform z-standardization removes each cohort's gene-level mean and variance *before* merging. This makes the joint expression matrix reflect within-platform biological variation rather than between-platform scale differences. After per-platform z-std, both LB and YFB recover non-zero $\hat{\beta}$ at $K = 3$.

4. Cohort Indicator: When It Helps vs. When It Hurts

The cohort indicator appends a platform dummy to L and estimates a cohort-specific gene profile F_{coh} via Normal conjugate update. Its effect depends on whether preprocessing has already removed the platform offset:

Setting	Cohort indicator effect	Mechanism
LB + joint QN (M1→M2)	+0.020	Absorbs platform offset; frees 2 extra biological factors
LB + per-platform z-std (M3→M4)	+0.004	Marginal; preprocessing already removed bulk offset
YFB + per-platform z-std (M5→M6)	−0.013	At $K = 3$, cohort column competes for variance with 2 biological factors
LB + joint QN, low K ($K=5$)	Essential	Without it, LB also collapses to $K_{\text{eff}} = 0$ at $K = 5$

Recommendation on cohort indicator: It is most valuable as a **robustness feature for LB when per-platform preprocessing cannot be guaranteed** (e.g., adding a new cohort to an existing fit without retraining). For the primary analysis configuration (YFB + per-platform z-std), it is *harmful* at low K and should be excluded. For sensitivity analysis (LB + per-platform z-std, M4), it is effectively neutral (+0.004) and can be included without cost. Future work extending to 3+ cohorts should revisit the cohort indicator, as it may provide greater benefit when platform contrast is larger.

5. Recommendation: Final Configuration for Manuscript

Table 3: Recommended configurations. $K=3$ is the biological floor (1-SE CV selected $K=2$ for YFB; $K=3$ has higher CV mean C and is consistent with DeSurv $K=3-4$).

							Mean	Notes
Role	Model	Preprocessing	Cohort	K	K_eff	C		
M5	Primary (M5)	YFB	Per-platform z-std	No	3	2	0.626	Exact external scoring; 2 active programs; highest C
M4	Sensitivity (M4)	LB	Per-platform z-std	Yes	3	1	0.624	Explicit platform offset model; LB robustness check

M5 (primary): YFB identifies 2 active gene expression programs ($K_{\text{eff}} = 2$) from 3 learned factors, with mean external $C = 0.626$ across 5 independent PDAC cohorts (Dijk, Moffitt GEO, PACA-AU array, PACA-AU seq, Puleo). New-patient risk scores require only $\eta_{\text{new}} = Y_{\text{new}} F \hat{\beta}$ — no sample-level latent variable approximation. Factor loadings F are directly interpretable as gene weights defining each prognostic program.

M4 (sensitivity): LB with cohort indicator reaches mean $C = 0.624$ — effectively tied with M5 — and explicitly encodes the RNA-seq vs. proteomics platform difference as a model parameter rather than a preprocessing step.

6. Summary Figure

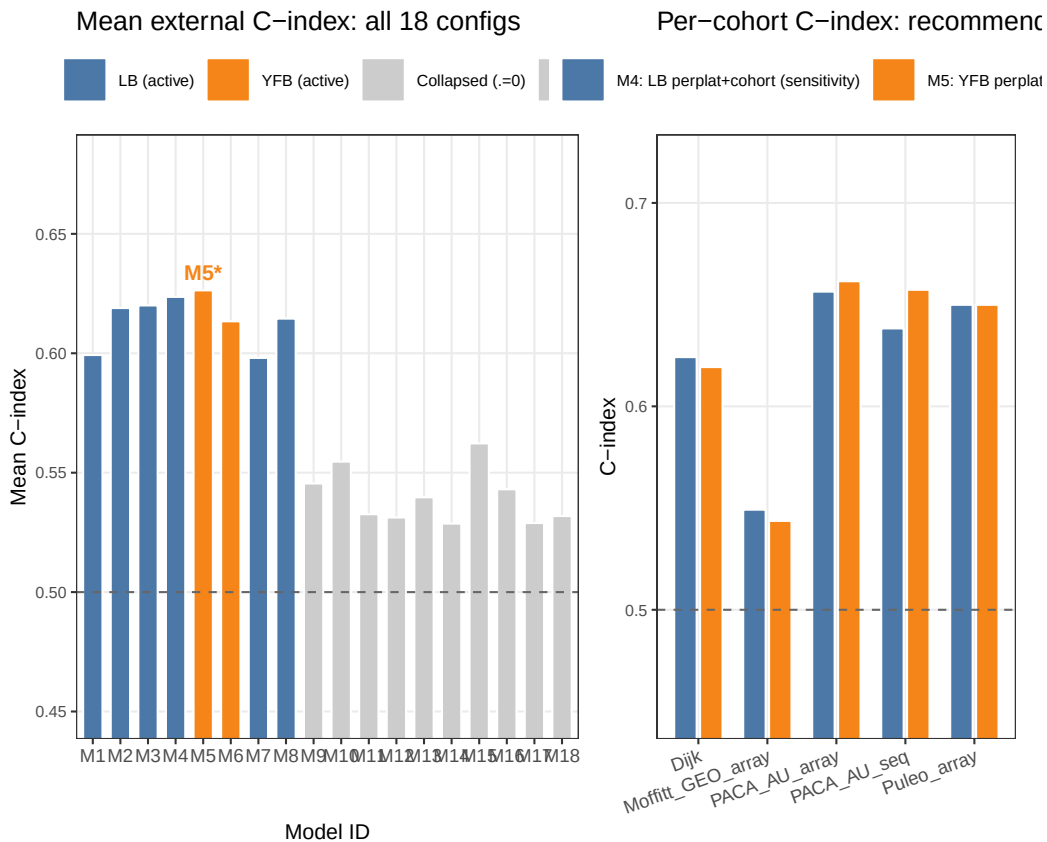


Figure 1: Left: mean external C-index for all 18 configurations, colored by model type. Dashed line = chance ($C=0.5$). Grey fill = configurations with $K_{\text{eff}}=0$ (beta collapse). Right: per-cohort C-index for the two recommended configurations.

7. Open Items and Next Steps

- **Gene set enrichment / pathway analysis** for M5 factors 1 and 2 (the 2 active prognostic programs). This is the primary interpretability task remaining before the manuscript methods section can be finalized.
- **Cohort indicator for future multi-cohort extensions.** As the training set grows beyond 2 platforms (e.g., adding microarray cohorts), the cohort indicator may become more valuable. The `cohort_id` parameter is already implemented in both LB and YFB; the cost/benefit analysis above applies per-pair of preprocessing choices and K .
- **Single-cohort YFB.** YFB on TCGA-only or CPTAC-only still has near-zero betas (lack of platform contrast means the SVD directions are more biological, but n is too small for stable CV). Not blocking for the merged-cohort manuscript.
- **LB prediction approximation.** The OLS approximation for LB new-subject scoring is adequate for sensitivity analysis but should be noted as a limitation vs. YFB. A potential alternative: train YFB on TCGA-only + CPTAC-only separately and compare factor consistency across platforms as a validation experiment.